

ASSIGNMENT



Inspiring Excellence

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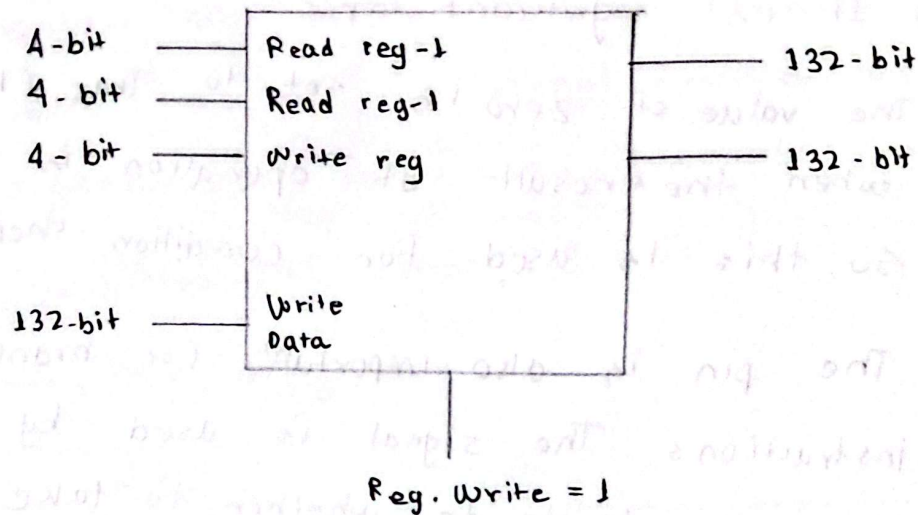
Section: 26

Assignment No: 4

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Course Code: CSE 340

Answer to the Question No. 1



Answer to the Question No. 2

For an Instruction [30, 14-12] in an ALU

Instruction [30] : Single bit in funct-7 field
to distinguish between operations
like ADD / SUB

Instruction [14-12] : Corresponds to 3-bit funct-3
field, specifying the core operation

Answer to the Question No. 3

The zero pin is an output signal from the ALU. It is significant for:-

(i) The value of zero is set to True (1) when the result of operation is 0. So, this is used for condition checking.

(ii) The pin is also important for branching instructions. The signal is used by datapath to decide whether to take branch or not.

Answer to the Question No. 4

For ADD, x_{21}, x_{22}, x_{23}

$$\begin{aligned}\text{Single Cycle Time} &= (IF + R.R + ALU + R.W) \\ &= (20 + 20 + 40 + 20)s \\ &= 100s\end{aligned}$$

$$\begin{aligned}\text{Pipeline Time} &= \text{no. of steps} \times \text{maximum time} \\ &= (500 \times 4)s \\ &= 2000s\end{aligned}$$

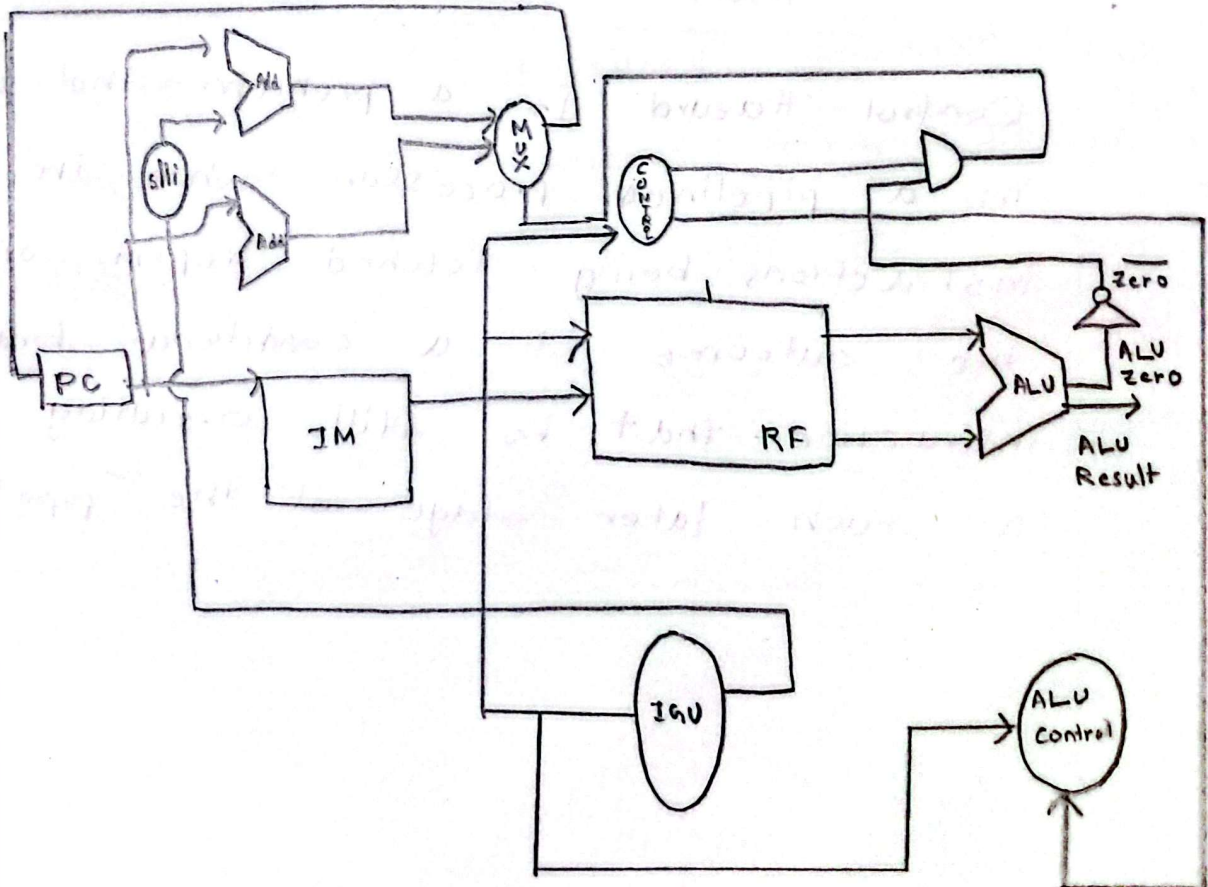
$\therefore$ Single cycle setup will run faster

Answer to the Question No 5

To implement BNE by DEQ, the branch should be taken if the registers is not zero. So, a not gate should be added to ALU zero pin. So, that,

$$PC_{src} \text{ BNE} = \text{Branch} \wedge \overline{\text{zero}}$$

So, the diagram:-



Answer to the Question No. 6

The required control signals will be: -

- (i) ALU Src = 1
- (ii) Reg Write = 1
- (iii) Mem Read = 1
- (iv) Mem Write = 0
- (v) Mem to Reg = 1

Answer to the Question No. 7

Control Hazard is a problem that arises in a pipelined processor when the instructions being fetched depends on the outcome of a conditional branch instruction that is still executing in a much later stage of the pipeline.

Answer to the Question No. 8

In the Harvard Architecture of RISC-V, instruction and data memory is kept separated to implement parallelism. Separating memory allows the processor to simultaneously fetch an instruction and read-write in the same clock cycle. In a pipelined processor, this separation of instruction fetch (IF) and Data Memory (DM) is very important.